Agentic Architecture Progression Dashboard

Report for Models 1–10

Alejandro Reynoso

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Abstract. This document presents a consolidated architectural dashboard for the ten chapters of AI for Financial Professionals Volume 4. The original wide-format tables have been reformulated into a more conventional academic structure by partitioning the comparison into two coherent blocks: Chapters 1–5 and Chapters 6–10. This substantially improves readability, page economy, and typographic balance. The report retains the original analytical purpose—to show the progression from elementary stateful workflows to more advanced agentic, adversarial, and supervisory architectures—while adopting a more restrained presentation consistent with a serious academic paper.

Scope. The report covers the full progression from Personal Finance Triage through Multi-Desk Research Synthesis. All chapters are assumed to operate in a LangGraph setting with bounded loops, explicit state transitions, and auditable outputs. Each workflow produces the standard audit bundle: `run_manifest.json`, `graph_spec.json`, and `final_state.json`.

1. Architecture Identity

Chapters 1–5

Dimension	CH1	CH2	CH3	CH4	CH5
Title	Personal Finance Triage	Suitability Gates	Credit Memo Agent	Hypothesis → Backtest Gate	Liquidity Regime Machine
Domain	Personal Finance	Investment Suitability	Credit Underwriting	Systematic Research	Execution / Microstructure
Core Question	What do I know and what is missing?	Is this request allowed?	Does the memo honour evidence discipline?	Does the strategy survive measured testing?	Are conditions safe enough to continue trading?
New Dimension	State, agents, and graph as the foundation	Refusal as first-class routing with hard termination	Structured critique loop with evidence-gap milestone	Deterministic tool replaces LLM-based measurement	Persistent regime state, hysteresis, and risk gate
Architecture Type	Conditional retry loop	Hard-branch gate with early termination	Critique–revision loop	Tool-augmented research gate	Stateful regime control loop
Loop Bound	`retry_count`	`consistency_passes`	`critique_max_iters`	`max_iters`	`max_steps`

Chapters 6–10

Dimension	CH6	CH7	CH8	CH9	CH10
Title	Rebalance Committee	Pitchbook Writing	M&A Due Diligence	Event-Driven Treasury	Multi-Desk Research
Domain	Portfolio Management	Investment Banking	M&A Advisory	Treasury Operations	Cross-Asset Research
Core Question	Do views converge to a feasible rebalance?	Has every section been drafted and gap-tagged?	What does the evidence say and what is missing?	What changed and what must we do now?	Is the cross-desk view ready for release?
New Dimension	Parallel committee with merge channels	Hub-and-spoke drafting queue	Domain routing, retrieval, and citation enforcement	Event-driven loop where time is primary	Multi-desk specialisation with red team and supervisor gate
Architecture Type	Parallel committee → aggregation → gate	Hub-and-spoke sequential drafting	Route → retrieve → answer → gap-check	Event-stream processing loop	Multi-agent pipeline → synthesis → adversarial review → supervisor gate
Loop Bound	`max_rounds`	`max_steps`	`max_loop_iters`	`max_steps`	`max_steps` + `max_revision_rounds`

2. Nodes and Roles

Chapters 1–5

Node Role	CH1	CH2	CH3	CH4	CH5
Entry / Intake	Intake (LLM)	Intake (LLM)	—	Hypothesis (LLM)	Market Data (det.)
Classification / Routing	Router (det.)	Suitability Gate	—	—	Regime Machine (det.)
Parallel Workers	—	—	—	—	—
Aggregation	—	—	—	—	—
Tool / Computation	—	—	—	Backtest Tool (det.)	Execution Tactic (det.)
LLM Output Node	Advisor / Clarify	Refusal / Guidance	Draft Memo / Revise Memo	Hypothesis / Review	Tactic Proposer
Adversarial Check	—	—	Critique Memo	Review	—
Hard Gate	—	Suitability Gate	Finalize	Router	Risk Gate
Milestone / State-only Node	—	—	Evidence Gaps	—	—

Chapters 6–10

Node Role	CH6	CH7	CH8	CH9	CH10
Entry / Intake	Intake (LLM)	Hub Router (det.)	Intake (det.)	Next Event (det.)	Intake (LLM/det.)
Classification / Routing	—	Hub Router (det.)	Router / Domain (LLM)	Covenant Gate (det.)	Desk Router (det.)
Parallel Workers	Committee Members ×3	Write Spokes $\times N$	—	—	Macro / Rates-FX / Equity / Credit Desks
Aggregation	Committee Reduce (det.)	Aggregate (det.)	—	—	Synthesis (LLM)
Tool / Computation	—	—	Retrieval (det.)	Apply Event + Metrics (det.)	—
LLM Output Node	Committee Members	Write Spokes	Answer Node	Notify	Desk Memos / Synthesis / Red Team
Adversarial Check	—	—	GapCheck	—	Red Team
Hard Gate	Gate Node	Hub Router	GapCheck Routing	Covenant Gate	Supervisor
Milestone / State-only Node	—	—	—	Advance (det.)	Revision Plan (det.)

3.LLM Role versus Deterministic Role

Chapters 1–5

Dimension	CH1	CH2	CH3	CH4	CH5
What the LLM does	Diagnoses missing information and drafts advice	Classifies request and drafts scoped response	Drafts, critiques, and revises memo	Proposes parameters and writes review	Proposes one tactic from bounded menu
What the LLM cannot do	Route or terminate	Override routing	Invent facts or self-approve	Compute backtest metrics	Control stop logic or regime state
What is deterministic	Routing and retry counter	Hard branch topology	Finalize and state milestones	Entire backtest measurement	Market evolution, execution simulation, and risk gate
Output contract	Text notes	Strict JSON profile	Strict JSON critique	Strict JSON hypothesis and review	Bounded tactic schema
Fallback if LLM fails	Retry loop	Consistency pass and simulator mode	One bounded repair	Deterministic baseline hypothesis	Deterministic tactic selection

Chapters 6–10

Dimension	CH6	CH7	CH8	CH9	CH10
What the LLM does	Analysts propose rebalances under mandates	Drafts assigned pitchbook sections	Classifies domain and drafts grounded answer	Drafts terse operating note	Desks write memos; synthesis integrates; red team challenges
What the LLM cannot do	Make final decision	Control section order	Route or invent citations	Compute liquidity metrics	Override supervisor gate
What is deterministic	Merge channels and enforce constraints	Queue management and assembly	Chunk scoring and citation filtering	Event application and covenant logic	Routing, revision planning, and artifact export
Output contract	Strict JSON proposals	Structured section template	JSON answer with citations	Short JSON-derived ops text	Structured JSON across all desk layers
Fallback if LLM fails	Error written to state	Node skipped with trace	Domain fallback	Placeholder note	Deterministic scaffold with error trace

4.Contracts and Enforcement

Chapters 1–5

Layer	CH1	CH2	CH3	CH4	CH5
Contract	Surface missing info before advising	Gate every request	Separate facts, assumptions, and open items	Measurement must come from code	Risk gate owns all stops
L1 Prompt	Classify completeness first	Stay within scope	Never fabricate facts	Return only bounded JSON	Choose only from allowed tactics
L2 Code Validation	`info_complete` check	Allowed decision values	Required keys and capped lists	Parameter clamping	Allowed tactic validation
L3 Policy Override	Retry cap	Refusal branch is terminal	Bounded iteration	Deterministic override of stop logic	Risk gate overrides everything
L4 Routing Reads State	Completeness and retries	Decision field	Memo quality and iterations	Decision and iteration count	Done flag only
Silent Failure Allowed	**No**	**No**	**No**	**No**	**No**

Chapters 6–10

Layer	CH6	CH7	CH8	CH9	CH10
Contract	Merge parallel proposals under constraints	Every section assigned and gap-tagged	Answer grounded only in retrieved evidence	Deterministic rules own financial actions	Multi-desk release must survive red-team challenge
L1 Prompt	One mandate per analyst	Use only the input pack	Use only evidence chunks	Return terse ops note only	Desk-specific prompts with explicit open items
L2 Code Validation	Merge-safe aggregation and clipping	Step count and section queue	Citation filtering	Hard-coded thresholds	Validation of required memo keys
L3 Policy Override	Gate reruns or escalates	Hard stop at max steps	Loop ends after bounded broadening	Max steps terminate event loop	Revision cap and reject-to-end routing
L4 Routing Reads State	Gate decision	Current section	Loop iterator	Covenant status	Pending desks and supervisor decision
Silent Failure Allowed	**No**	**No**	**No**	**No**	**No**

5.Loop Mechanics

Chapters 1–5

Dimension	CH1	CH2	CH3	CH4	CH5
What triggers iteration	Missing info with retries remaining	Inconsistent suitability decision	Revise flag with iterations remaining	Iterate decision with remaining count	Workflow not done and steps remain
What terminates	Completeness or retry limit	Stable decision or pass exhaustion	Pass or max iterations	Stop or iteration limit	Breach, completion, or max steps
Who owns the stop	Router	Suitability Gate	Finalize Node	Router	Risk Gate
Terminal statuses	Advice or stop	Allow / Refuse / More Info / Human Review	Finalized / Stopped	Stop with measured metrics	Order Complete / Breach / Max Steps
Loop visible in graph	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

Chapters 6–10

Dimension	CH6	CH7	CH8	CH9	CH10
What triggers iteration	High disagreement with rounds remaining	Pending sections with steps remaining	Empty citations or open items	Remaining events and steps	Revise decision with revision rounds left
What terminates	Approval, rejection, review, or max rounds	Sections complete or max steps	Evidence sufficient or loop limit	No events or max steps	Approve, reject, or bounded revision exhaustion
Who owns the stop	Gate Node	Hub Router	GapCheck Routing	Next Event Routing	Supervisor Routing
Terminal statuses	Approved / Rejected / Human Review	Completed / Max Steps	Evidence OK / Max Iters / Escalated	Stream Exhausted / Max Steps	Approved / Rejected / Revision Cap / Max Steps
Loop visible in graph	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

6.State Schema Evolution

Chapters 1–5

Field Category	CH1	CH2	CH3	CH4	CH5
Run Identity	run id, timestamp, model	run id, config hash, environment	run id, model, started utc	run id, config hash	run id, start utc, model
Input	user input	raw request, extracted profile	case facts	user request	order id, quantity, urgency
Parallel Fields	—	—	—	—	—
Working Outputs	notes, final output	suitability decision, response	memo, critique, evidence gaps	hypothesis, backtest, review	tactic, fills, remaining qty
Control Fields	completeness, retries	terminated, consistency passes	iteration count, max iters, status	iteration count, decision	step, max steps, done
Domain Fields	—	—	—	—	prices, spread, depth, regime, risk metrics
LLM Tracking	—	—	—	hypothesis errors	llm calls, confidence, rationale, errors
Audit Trail	basic trace	trace	trace	bounded trace	trace, notes, errors

Chapters 6–10

Field Category	CH6	CH7	CH8	CH9	CH10
Run Identity	run id, config, versions	run id, timestamp, model	run id, timestamp, config	run id, timestamp, hash	run id, timestamp, config hash
Input	portfolio, constraints, snapshot	input pack, pending sections	question	event queue, event index	query, constraints
Parallel Fields	committee reports, merge-safe errors	section outputs, section gaps	—	—	desk memos, pending desks, completed desks
Working Outputs	proposed trades, committee scores, decision	assembled pitchbook, section outputs	domain, retrieved chunks, answer	last event, action, alerts, ops note	synthesis, red-team critique, supervisor decision
Control Fields	round, max rounds, gate decision	steps, max steps, current section	loop iter, max loop iters	step, max steps	steps, revision rounds, max revision rounds

Field Category	CH6	CH7	CH8	CH9	CH10
Domain Fields	portfolio weights and ticker features	—	document-type restrictions	liquidity, utilization, covenant status	—
LLM Tracking	merge-safe errors	refusal and reason	—	llm status per alert	parse errors and trace
Audit Trail	merge-safe trace	trace	trace	node-level trace	trace and errors

7.Final Deliverables

Chapters 1–5

Artifact	CH1	CH2	CH3	CH4	CH5
run_manifest.json	run id, timestamp, model	run id, config, hash, env	run id, model lock, bounds	run id, model lock, outputs summary	run id, config hash, llm usage, outcome
graph_spec.json	nodes, edges, mermaid	routing decisions and mermaid	conditional routing	loop bound and mermaid	topology and mermaid
final_state.json	full state	profile, decision, response	memo, critique, gaps, assumptions	hypothesis, backtest, review, decision	execution state, regime, llm proposals, trace
Extra Deliverable	—	—	—	—	—
Primary Human Deliverable	Advice or clarifying questions	Scoped response or refusal	Credit memo plus open items	Hypothesis, results, and decision	Execution log and stop reason

Chapters 6–10

Artifact	CH6	CH7	CH8	CH9	CH10
run_manifest.json	config, versions, round summary	model lock, controls, topology tag	question hash and outcome	steps and alerts	zip plus SHA-256
graph_spec.json	merge channels and loop bounds	hub-and-spoke topology	retrieval method and loop bounds	architecture notes	desk list and supervisor topology
final_state.json	portfolio, proposals, scores, decision	section outputs and assembled pitchbook	chunks, answer, citations, open items	liquidity state, alerts, ops notes	desk memos, synthesis, critique, supervisor decision
Extra Deliverable	—	`research_report.txt`		—	`research_report.txt` + ZIP
Primary Human Deliverable	Proposed rebalance and disagreement score	Assembled pitchbook and gap list	Structured diligence answer	Event log and liquidity position	Multi-desk research packet

8.Governance Feature Matrix

Chapters 1–5

Feature	CH1	CH2	CH3	CH4	CH5
State-driven routing	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Bounded loop	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Explicit end node	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Strict JSON contract from LLM	**No**	**Yes**	**Yes**	**Yes**	**Yes**
Hard refusal / early termination	**No**	**Yes**	**No**	**No**	**Yes**
Structured critique / adversarial check	**No**	**No**	**Yes**	**No**	**No**
Deterministic tool computation	**No**	**No**	**No**	**Yes**	**Yes**
LLM overridden by deterministic policy	**No**	**Yes**	**Yes**	**Yes**	**Yes**
Deterministic fallback if LLM fails	**No**	**Yes**	**Yes**	**Yes**	**Yes**
Parallel agents / multi-perspective	**No**	**No**	**No**	**No**	**No**
Merge channels	**No**	**No**	**No**	**No**	**No**
Hub-and-spoke topology	**No**	**No**	**No**	**No**	**No**
Retrieval / evidence grounding	**No**	**No**	**No**	**No**	**No**
Domain / intent routing	**No**	**Yes**	**No**	**No**	**No**
Event-driven loop	**No**	**No**	**No**	**No**	**Yes**
Persistent regime / operational state	**No**	**No**	**No**	**No**	**Yes**
Hard risk / cost gate	**No**	**Yes**	**No**	**No**	**Yes**
Supervisor / approval gate	**No**	**Yes**	**No**	**No**	**No**
LLM action space bounded to menu	**No**	**No**	**No**	**Yes**	**Yes**
LLM usage tracked in state	**No**	**No**	**No**	**No**	**Yes**
Artifact audit bundle	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Disagreement / confidence scoring	**No**	**No**	**No**	**No**	**No**

Chapters 6–10

Feature	CH6	CH7	CH8	CH9	CH10
State-driven routing	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Bounded loop	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Explicit end node	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Strict JSON contract from LLM	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Hard refusal / early termination	**Yes**	**No**	**No**	**No**	**Yes**
Structured critique / adversarial check	**No**	**No**	**Yes**	**No**	**Yes**
Deterministic tool computation	**Yes**	**No**	**Yes**	**Yes**	**No**
LLM overridden by deterministic policy	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Deterministic fallback if LLM fails	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Parallel agents / multi-perspective	**Yes**	**No**	**No**	**No**	**Yes**
Merge channels	**Yes**	**No**	**No**	**No**	**No**
Hub-and-spoke topology	**No**	**Yes**	**No**	**No**	**Yes**
Retrieval / evidence grounding	**No**	**No**	**Yes**	**No**	**No**
Domain / intent routing	**No**	**No**	**Yes**	**No**	**Yes**
Event-driven loop	**No**	**No**	**No**	**Yes**	**No**
Persistent regime / operational state	**No**	**No**	**No**	**Yes**	**No**
Hard risk / cost gate	**Yes**	**No**	**No**	**Yes**	**No**
Supervisor / approval gate	**Yes**	**No**	**No**	**No**	**Yes**
LLM action space bounded to menu	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
LLM usage tracked in state	**No**	**No**	**No**	**Yes**	**Yes**
Artifact audit bundle	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Disagreement / confidence scoring	**Yes**	**No**	**No**	**No**	**No**

9.Architectural Progression Narrative

#	Chapter	New Capability Added	Analogy in Real Finance
1	Personal Finance Triage	Introduces state, agents, and graph as primitives, with a simple retry loop for missing information.	An analyst asking clarifying questions before beginning substantive work.
2	Suitability Gates	Adds hard branching, refusal, and early termination as legitimate outputs of the system.	A compliance gate that can block a recommendation before it reaches a client.
3	Credit Memo Agent	Introduces a structured critique loop and treats evidence gaps as named state milestones.	A credit committee returning a memorandum for revision before approval.
4	Hypothesis → Backtest Gate	Replaces LLM self-evaluation with deterministic measurement and policy-driven stopping.	A research team running a measured backtest before escalating a trading idea.
5	Liquidity Regime Machine	Adds persistent market state, hysteresis, and a hard risk gate governing workflow termination.	An execution desk pausing activity when liquidity conditions deteriorate.
6	Rebalance Committee	Adds parallel deliberation, merge-safe state updates, and disagreement as a governance signal.	An investment committee whose analysts work in parallel before portfolio manager review.
7	Pitchbook Writing	Moves to hub-and-spoke architecture, with a sequential section queue and bounded drafting.	A production line for pitchbook preparation with coordinator and section writers.
8	M&A Due Diligence	Adds routing by domain, retrieval, citation enforcement, and an evidence sufficiency loop.	A diligence analyst assembling a memo only from retrieved documents and explicit open items.
9	Event-Driven Treasury	Makes time and event flow, rather than user prompting, the driver of the graph.	A treasury monitoring system processing feeds and issuing operating alerts.
10	Multi-Desk Research	Introduces desk specialisation, adversarial review, and supervisory approval with bounded revision.	A cross-desk research workflow ending in CIO-level release control.

10. One-Line Summary per Chapter

#	Chapter	Domain	What the Loop Enforces
1	Personal Finance Triage	Personal Finance	Missing information must be surfaced before advice is produced.
2	Suitability Gates	Investment Suitability	Every request must be accepted, refused, or escalated before any substantive response.
3	Credit Memo Agent	Credit Underwriting	Facts, assumptions, and open items must remain explicitly separated.
4	Hypothesis → Backtest	Systematic Research	Performance claims must originate in deterministic computation rather than LLM narration.
5	Liquidity Regime Machine	Execution / Microstructure	Trading continues only while deterministic risk conditions remain acceptable.
6	Rebalance Committee	Portfolio Management	Parallel proposals must converge under constraints or escalate on disagreement.
7	Pitchbook Writing	Investment Banking	Every section must be assigned, drafted, and gap-tagged before assembly.
8	M&A Due Diligence	M&A Advisory	Every answer must be grounded in retrieved evidence and explicit citations.
9	Event-Driven Treasury	Treasury Operations	Each material event must be processed and actioned under deterministic operating policy.
10	Multi-Desk Research	Cross-Asset Research	Cross-desk coverage must survive adversarial review and supervisor approval before release.

Conclusion

The revised structure replaces extremely wide dashboard tables with a two-block academic presentation that is considerably easier to read, print, and maintain. The underlying conceptual progression remains unchanged: the early chapters establish stateful and bounded graph primitives; the middle chapters introduce deterministic validation, persistence, and committee structures; and the final chapters move toward retrieval-grounded, event-driven, and supervisor-controlled architectures. Partitioning the material into Chapters 1–5 and Chapters 6–10 is not merely a formatting repair. It also reflects a substantive pedagogical divide between foundational and advanced agentic forms.